



Python, pip, Django & Virtual Env Setup (Lecture #3)

Installation, PATH Setup, Package Management, Isolation & VS Code Integration

Master Class Notes • Mohit Decodes Playlist #3 • Hinglish Quick Revision

🎯 Welcome Soubhik! (Personal Memory & Context Link):

Aapne Windows PowerShell me Python environment aur multiple Python installations (jaise Python 3.11/3.14 pathing) par kaam kiya hai. Yahi problem Django ke real-world projects me aati hai: agar system me global packages daalenge toh versions aapas me takraate (conflict) hain. Isi wajah se **Virtual Environment (venv)** setup karna professional Django developers ka pehla aur sabse zaroori kadam hota hai!



1. Complete Setup Pipeline (Bird's-Eye View)



2. Step 1: Python Installation & Version Check

Django run karne ke liye recommended hai ki aapke paas **Python 3.8+** hona chahiye (purane versions end-of-life ho chuke hain).



Windows Installation Golden Rule:

Python installer run karte waqt sabse niche aane wale checkbox: ☒ Add python.exe to PATH ko zaroor check karein! Agar ye skip ho gaya toh terminal me *'python' is not recognized* ka error aayega.

Terminal / PowerShell me check karne ke liye:

```
# Windows (CMD / PowerShell):  
python --version  
  
# macOS / Linux:  
python3 --version
```



3. Step 2: pip (Python Package Manager)

pip Python ka default package installer hai. Jaise Android me Google Play Store ya iPhone me App Store hota hai jahan se nayi apps download hoti hain, waise hi Python me third-party libraries (jaise Django, Flask, Pandas) ko download aur install karne ka kaam **pip** karta hai.

pip Python ke sath hi automatically install hota hai. Verify karne ke commands:

```
# Windows:
pip --version

# macOS / Linux:
pip3 --version
```

4. Step 3: Virtual Environment (Kyun Zaroori Hai?)

Virtual Environment ek **isolated room (container)** ki tarah hota hai jo aapke system ke global Python ko chhue bina har project ke liye alag versions aur packages manage karta hai.

Real-Life Analogy (Alag-Alag Bartan):

Maan lijiye aapke pass do orders hain:

- **Project A (Old Client):** Isko purani recipe (Django 4.2 aur Python 3.10) chahiye.
- **Project B (New Lighting Store App):** Isko latest recipe (Django 5.2 aur Python 3.13) chahiye.

Agar aap dono ko ek hi bartan (Global System) me mix kar denge toh sab kharab (conflict) ho jayega! Virtual Environment har project ko apna alag, clean tiffin box deta hai.

Virtual Environment Setup Commands (Step-by-Step):

Action	Windows (PowerShell / CMD)	macOS / Linux
1. Install virtualenv tool	<code>pip install virtualenv</code>	<code>pip3 install virtualenv</code>
2. Create Environment	<code>virtualenv my_env</code>	<code>virtualenv my_env</code>
3. Activate Environment	<code>my_env\Scripts ctivate</code>	<code>source my_env/bin/activate</code>
4. Deactivate (Exit)	<code>deactivate</code>	<code>deactivate</code>

Success Indication: Jab environment successfully activate ho jata hai, tab terminal prompt ke shuru me bracket me naam dikhta hai, jaise: `(my_env) PS C:\...>`

5. Step 4: Django Installation & Verification

Jab aapka virtual environment activate ho jaye `(my_env)`, tabhi Django install karein taaki wo global system me clutter na failaye:

```
# 1. Install Django (Virtual environment active hona chahiye):
pip install django

# 2. Check Django Version:
django-admin --version
```

VS Code Tip:

VS Code me folder open karein (e.g. `django_project`) aur **Ctrl + ` (Backtick)** daba kar VS Code ka integrated terminal open kar lein. Wahan se terminal commands aur files dono ek sath ek hi window me manage ho jaate hain.



6. Top 5 Interview / Exam Quick Revision Q&A

1. Q: Django ke liye minimum kaun sa Python version recommended hai?

Ans: Python 3.8+ (kyunki 3.8 se pehle ke versions officially end-of-life ho chuke hain).

2. Q: pip ka full form aur primary function kya hai?

Ans: **pip** Python ka package manager hai jo PyPI (Python Package Index) se external packages aur libraries ko install/manage karta hai.

3. Q: Virtual Environment banana kyun recommended mana jata hai?

Ans: Dependency isolation ke liye — taaki alag-alag projects ke library versions aapas me conflict na karein aur global Python system clean rahe.

4. Q: Windows par virtual environment activate karne ka command kya hai?

Ans: `my_env\Scripts activate` (aur exit karne ke liye `deactivate`).

5. Q: Django version check karne ke liye CLI tool kaun sa use hota hai?

Ans: `django-admin --version` ya Python shell me `import django; print(django.__version__)` .